

Building Systems for Scalable and Responsible Data Management

The Data Intelligence Loop

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Motivation

My research asks how to build data systems that scale to complex workloads, support reliable AI workflows over data, and inform responsible real-world decision-making.

Complex and dynamic data

Tables, graphs, text, documents, multimedia data, logs, streams, and evolving workloads.

AI data workloads

Retrieval, indexing, evidence selection, reasoning, verification, serving, and cost control.

Real-world applications

ESG, finance, workforce analytics, fairness, climate, health, and scientific discovery.

Database systems provide the foundation for managing, processing, and reasoning over data across these motivations.

Three-Theme Research Agenda

STAGE



Data Processing

Process heterogeneous and dynamic data efficiently.



Data Understanding

Extract structure, semantics, context, and evidence.



Data Intelligence

Support responsible decisions in real domains.

RESEARCH FOCUS

THEME 1

Scalable Data Systems

Graphs, tables, streams, query processing, GPU, out-of-core

THEME 2

Responsible Data Intelligence

GraphRAG, LLM reasoning, AI/ML for DB, tabular learning, data-centric AI

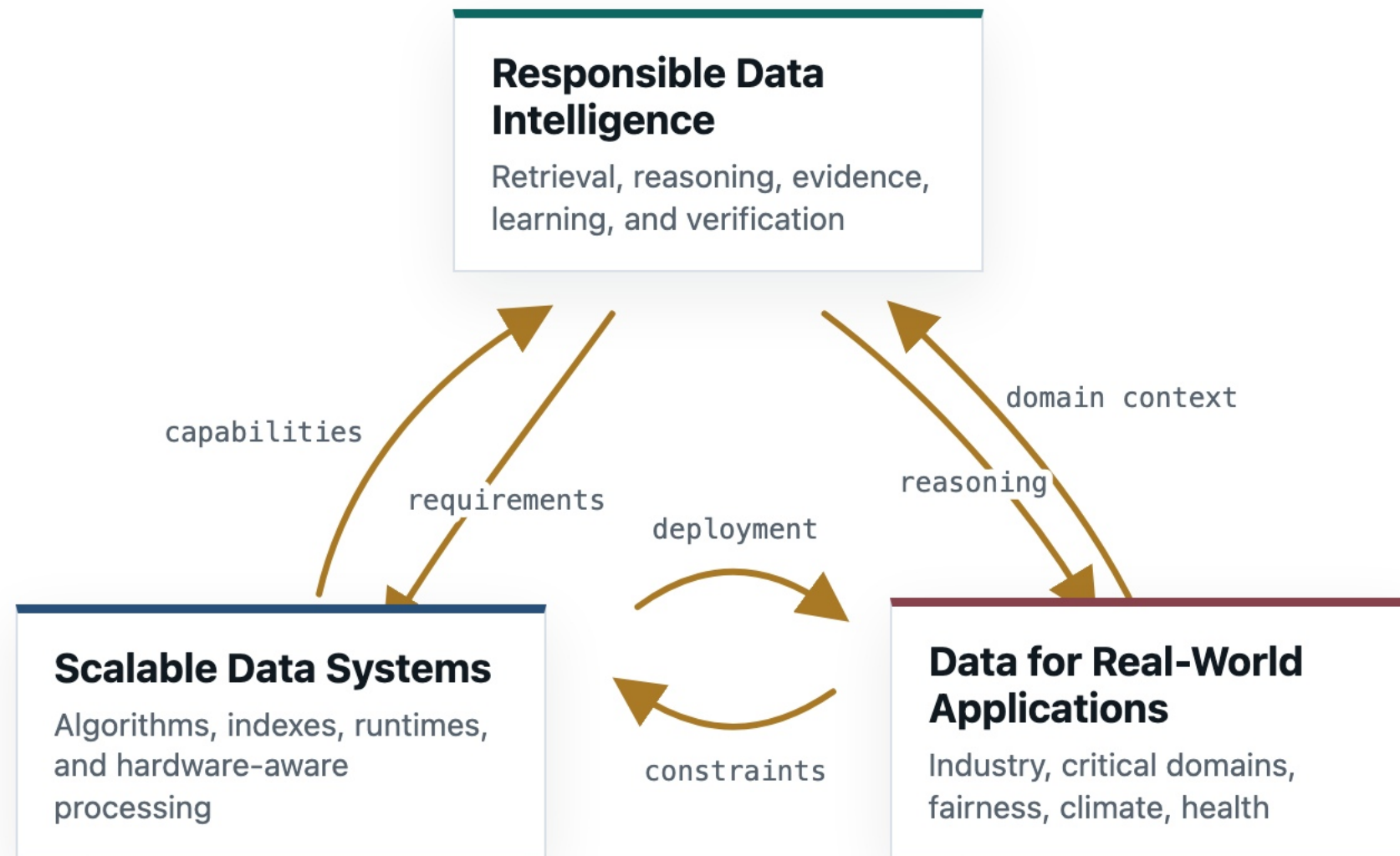
THEME 3

Data for Real-World Applications

Industry collaboration, critical domains, fairness, climate, health

The agenda links scalable systems, responsible intelligence, and real-world applications through three themes.

The Data Intelligence Loop



BOTTOM-UP

Systems enable intelligence

Efficient data systems organize and process complex data, so AI methods can retrieve evidence, reason over it, and support decisions.

TOP-DOWN

Applications define requirements

Real-world use cases tell us what the systems must guarantee: traceable evidence, reliable outputs, fair decisions, validation, and deployment readiness.

Processing enables understanding; understanding enables intelligence; real-world applications feed requirements back into data systems.

PART I

Scalable Data Systems

Data processing is about making complex and dynamic data computationally manageable at scale.

Graph Analytics and Processing

Graphs, hypergraphs, temporal graphs, matching, centrality, decomposition.

Database Systems & Query Processing

Query processing, joins, JIT/runtime, OLTP kernels, storage, transactions.

Parallel and Hardware-Aware Processing

GPU, FPGA, RISC-V, UMA, streaming, parallelism, memory- and I/O-aware execution.

Graph Analytics at Scale

Complex relational structures are no longer only graphs; they include higher-order and temporal interactions that make naive enumeration infeasible.

REPRESENTATIVE TOPICS

- **Higher-order graph and hypergraph** analytics
- **Pattern matching** and **substructure search**
- **Connectivity and path** analytics
- **Temporal and evolving graph** analytics

Turn structural properties into indexes, pruning rules, and execution strategies that avoid exhaustive search.

SELECTED WORK

COMPLETED

Nucleus decomposition

SIGMOD 2026

Diversified graph pattern matching

VLDB 2026

Hypergraph matching

ICDE 2026

ICDE 2023

Hypergraph decomposition

VLDBJ 2026

SIGMOD 2025

VLDBJ 2025

CIKM 2024

Constrained path analytics

ICDE 2025

Clique and dense subgraph analytics

WWW 2025

Temporal graph analytics

WWW 2024

Graph pattern matching systems

SIGMOD 2021

ICDE 2021

VLDB 2019

UNDER SUBMISSION

Higher-order graph analytics

SPRINGER MONOGRAPH, IN PRESS

Nucleus and nuclear decomposition

SIGMOD 2027

VLDB 2027

Hypergraph core maintenance

ICDE 2027

Reachability indexing and querying

EDBT 2026 UNDER REVISION

Database Systems & Query Processing

Database performance depends not only on algorithms but also on how query logic interacts with compilation, runtime execution, storage, and deployment constraints.

REPRESENTATIVE TOPICS

- **Transactional engine** design
- **Query processing**, joins, and **runtime optimization**
- **JIT / MLIR** compilation for **database execution**
- **Storage, concurrency**, and **recovery**

Systems optimization should make database engines faster while keeping them maintainable and deployable.

EXECUTION PIPELINE

01

Query

Declarative intent and workload shape.



02

Runtime

JIT, MLIR, and specialized execution.



03

Engine

Storage, transactions, and deployment.

SELECTED WORK

SPOTLIGHT

PhoebeDB: High-Performance Disk-Based RDBMS Kernel for OLTP

EDBT 2025 INDUSTRY INDUSTRY COLLABORATION TARGET: INDUSTRY-READY

- **Industry collaboration** toward an **enterprise-oriented**, industry-ready RDBMS kernel.
- **PostgreSQL-compatible OLTP** on **commodity hardware**, reducing migration and deployment risk.
- **Full-stack kernel design**: temperature-aware storage, coroutine runtime, smart scheduling, transaction management, WAL, and snapshots.
- Reports **13.7M tpmC** and **30M tpm** on **TPC-C** using one machine, with **27x** improvement over PostgreSQL.

COMPLETED

Batch-dynamic high-dim kNN join

ADMA 2024 BEST STUDENT PAPER

UNDER SUBMISSION

Fully dynamic high-dim kNN joins

VLDBJ

Parallel and Hardware-Aware Processing

At billion scale, performance depends on coordinating distributed execution, parallelism, memory hierarchy, storage, and accelerators.

REPRESENTATIVE TOPICS

- **Distributed** and out-of-core processing
- **Parallel runtime** and **adaptive scheduling**
- **GPU-/FPGA-accelerated indexing** and analytics
- **Emerging hardware** and **scalable execution**

Scale by co-designing algorithms, runtimes, and hardware-aware execution.

EXECUTION DIMENSIONS

Distributed execution

Memory / Out-of-core

Parallel runtimes

GPU / FPGA / RISC-V

SELECTED WORK

COMPLETED

Out-of-core graph processing system

SIGMOD 2026

GPU-accelerated vector index construction

SIGMOD 2026

GPU-accelerated range top-k query

HPDC 2026

Database benchmarking on RISC-V

VLDB ADMS 2025

UNDER SUBMISSION

FPGA-accelerated graph random walks

SIGMOD 2027

SIMD set-intersection benchmarking

SIGMOD 2027

Out-of-core vector search and indexing

EUROSYS 2027

Unified-memory DataFrame processing

VLDB 2026 DEMO

PART II

Responsible Data Intelligence

Data understanding is about turning processed data into structure, semantics, context, and reliable evidence for AI systems.

Data Systems for Reliable LLMs

Retrieval, evidence selection, reasoning, agent memory, and context management.

Learning-Augmented Data Systems

Cost estimation, tuning, temporal analytics, and costly system decisions.

Data-Centric AI

Relational, unstructured, and graph data; efficient learning.

Data Systems for Reliable LLMs

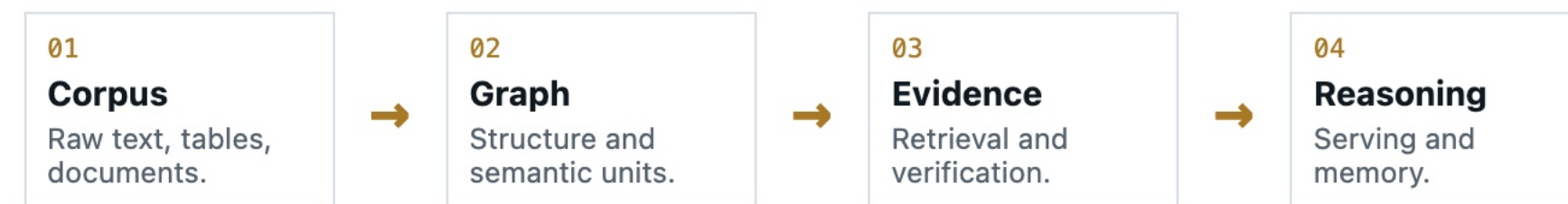
Reliable LLM applications over data need systems for retrieval, evidence organization, verification, reasoning orchestration, and serving, not just fluent generation.

REPRESENTATIVE TOPICS

- **Data-to-structure** construction
- **Evidence retrieval** and **selection**
- **Reasoning** and **verification**
- **Serving, memory**, and **context management**
- **LLM safety** and **jailbreaking**

Reliable LLMs create data-management workloads: indexing, planning, provenance, cost control, and evidence quality.

RETRIEVAL-TO-REASONING PIPELINE



SELECTED WORK

● COMPLETED

Adaptive GraphRAG retrieval

KDEXLLM 2026 BEST PAPER

LLM-graph for data exploration

SIGMOD DEMO 2025

● UNDER SUBMISSION

Graph CoT reasoning and serving

TOIS

Cost-efficient KG construction

TKDE

Semantic-unit retrieval for HyperRAG

EMNLP 2026

Long-term memory for LLM agents

EMNLP 2026

Memory-driven LLM jailbreaking

EMNLP 2026

Learning-Augmented Data Systems

This line of work learns to approximate, tune, and adapt costly decisions in data systems while respecting query, graph, and temporal semantics.

REPRESENTATIVE TOPICS

- **Learning** for **query optimization**
- **Graph neural networks (GNNs)**
- **Representation learning** over **structured data**
- **Deployment-aware** learned components

AI/ML is useful when learning acts as a system component with clear semantics, error behavior, and operational cost.

Query Optimization

Cost estimation and optimization signals for expensive database decisions.

Graph Analytics

Temporal centrality and graph signals where exact computation is costly.

SELECTED WORK

● COMPLETED

Learning-based temporal centrality

WWW 2026

WWW 2024

Adaptive parallelism tuning for stream processing

ICDE 2025

Heterogeneous graph learning

CIKM 2025

● UNDER SUBMISSION

Learned query cost estimation

ICDE 2026

Event-centric HIN learning

ICDM 2026

Learning-based attributed community detection

TKDD

Data-Centric AI

Data-centric AI studies how relational, unstructured, and graph data shape model learning, reasoning, efficiency, and reliability.

REPRESENTATIVE TOPICS

- **Structured data representation** and **reasoning**
- **Data condensation** and **efficiency**
- **Label quality** and **aggregation**
- **Data quality** for **AI workflows**

Data-centric AI brings classic data quality questions back into the core of model performance and reasoning reliability.

DATA PROBLEMS

Relational data

Tables, schemas, joins, and structured reasoning.

Unstructured data

Text, documents, and multimodal evidence.

Graph data

Entities, relations, paths, and context.

Efficiency

Less data, lower cost, and faster learning.

SELECTED WORK

● COMPLETED

Tabular data condensation

ICDE 2026

Structured table reasoning

WWWJ 2025

● UNDER SUBMISSION

LLM-based tabular reasoning

EMNLP 2026

Label aggregation in crowdsourcing

ICDE 2026

PART III

Data for Real-World Applications

Data intelligence is about using processed and interpreted data to support responsible decisions in real-world domains.

Industry Collaboration

Critical domains: ESG, finance, law, mining, energy, power, and built environment.

Fair Decision Support

Representation, bias, evaluation, context.

Science and Society

Climate, health, social/economic simulation, materials.

Industry-Critical Applications

Industry collaborations in critical domains such as ESG, finance, law, mining, energy, and built environment turn messy records into decisions that require traceability, interpretation, and accountability.

REPRESENTATIVE TOPICS

- **ESG, finance, and sustainability** evidence governance
- **Built environment and infrastructure** decision support
- **Mining, resources, and environmental** compliance
- **Energy, solar, and power-system** analytics
- **Law, regulation, and professional** decision support

Industry collaboration turns real operational needs into data-system problems: entities, evidence, standards, provenance, and auditable workflows.

SELECTED WORK

COMPLETED

Agent-based ESG report analysis

ICSA 2026

Finance analysis and fraud detection

BIGCOMP 2026 ADMA 2025

Construction career guidance

BIGCOMP 2026 ISARC 2026

Built-environment GraphRAG systems

BUILDINGS 2026

Workforce progression analysis

BUILDINGS 2026 CRIOCM 2026 IJCM 2025

UNDER SUBMISSION

ESG data management survey

THE INNOVATION

Standard-aware ESG disclosure analysis

ICDM DEMO 2026

Mining ESG analysis

APWEB-WAIM

ESG knowledge graph construction

ICKG 2026

Built-environment AI co-pilot

AUTOMATION IN CONSTRUCTION

Compliance and sustainability intelligence

ARC LINKAGE PROJECT

ESG Data Management

AI-Empowered ESG Data Analytics in Responsible Investment

The project leverages advanced AI and data management to improve ESG risk assessment, performance forecasting, and report generation.

BACKGROUND

ESG links environmental impact, social responsibility, and governance quality to investment, compliance, and accountability.

SIGNIFICANCE

Reliable data infrastructure makes ESG claims traceable, comparable, and actionable for responsible investment.

FOR 4605 - Data Management and Data Science

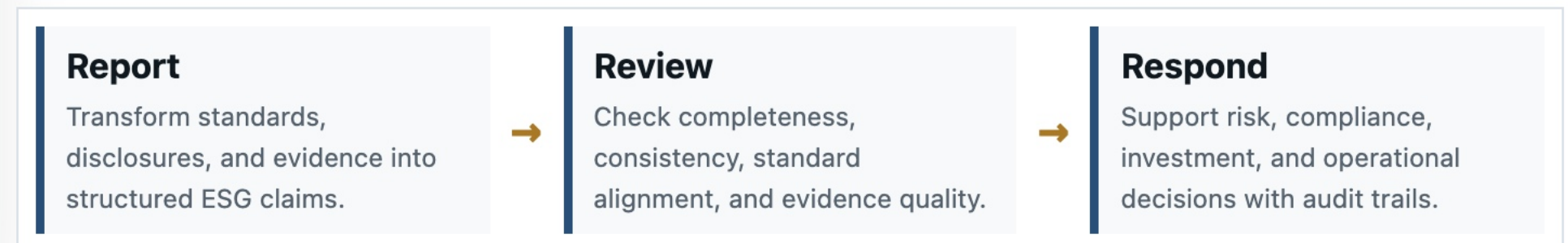
ESG analysis is a data-management problem before it is an AI problem: fragmented standards, disclosures, market signals, enterprise evidence, risk signals, forecasts, and reports must become timely, objective, and auditable decision support.

PROJECT AIMS	Assess	Forecast	Generate
	Improve ESG risk analysis from complex, mixed-source data.	Surface performance signals for responsible investment.	Produce transparent reports backed by auditable evidence.

NEED -> OUTPUT -> DATA -> SYSTEM

National need Reliable ESG insights support sustainable investment, corporate responsibility, and regulatory compliance.	Expected outcome More accurate, accessible, timely, and objective ESG analysis for investors and companies.
Data science challenges Availability, quality, integration, traceability, timeliness, privacy, and cross-standard semantics.	Systems agenda ESG knowledge layer, claim-level provenance, retrieval/reasoning guardrails, and workflow evaluation.

GOVERNED ESG DECISION LOOP



Data Foundations for Fair Decisions

Fairness work requires careful data modeling because representation, missingness, and measurement choices shape downstream conclusions.

REPRESENTATIVE TOPICS

- **Representation** and **measurement**
- **Fairness** in decision support
- **Diversity and progression** analytics
- **Evaluation** of **model behavior**

Fairness is not only a model metric; it is also a data-management problem of measurement, provenance, and context.

FAIRNESS QUESTIONS

Who is underrepresented?

Groups missing or poorly measured.

Who is affected?

Real decisions shaped by outputs.

What harm can scale?

Bias amplified across workflows.

What is the social impact?

Fairness, opportunity, and trust.

SELECTED WORK

● COMPLETED

Gender representation and career progression

JCEM 2026 IJCM 2025 IJCM 2025

LLM fairness and bias evaluation

FAIML 2026 AI 2025

● UNDER SUBMISSION

LLM bias in hiring

WORK IN PROGRESS

Data Workloads in Science and Society

Scientific and societal applications need AI pipelines that respect domain constraints and produce outputs that can be validated.

REPRESENTATIVE TOPICS

- **Domain-aware** scientific modeling
- **Health and conversation** intelligence
- **Social simulation** and **economic simulation**
- **Scientific discovery** workflows

Domain-facing intelligence needs data systems that make heterogeneous evidence scalable, checkable, and reusable.

DOMAIN PILLARS

Science

Climate modeling, materials discovery, and validation-centered AI.

Social

Social simulation, economic simulation, and policy-facing analysis.

Health

Conversation data, semantics, and responsible interpretation.

SELECTED WORK

● UNDER SUBMISSION

Weather and climate modeling

NEURIPS 2026

Health conversation data intelligence

ADMA 2026

AI-assisted materials discovery

MULTIPLE MANUSCRIPTS

What This Agenda Enables

A database systems agenda for AI-era decision-making.

From data systems to decision support.

The common thread is to turn complex data into scalable computation, reliable understanding, and responsible decision support in real domains.

SYSTEMS

AI PIPELINES

APPLICATIONS

CAPABILITY

Scalable data substrates

Algorithms, indexes, runtimes, and hardware-aware execution for complex workloads.

RELIABILITY

AI-facing data understanding

Retrieval, reasoning, provenance, and verification over structured and unstructured data.

IMPACT

Domain-grounded intelligence

Real-world settings feed back requirements for fairness, robustness, auditability, and deployment.

The broader contribution is to connect database systems, AI pipelines, and real-world data deployment through one coherent data intelligence loop.

Future Directions

DIRECTION 1

Data Systems for Reliable LLMs

- RAG-aware indexing and retrieval planning
- Evidence-aware query processing and verification
- Cost-based optimization for reasoning pipelines
- Provenance and audit trails for LLM outputs

DIRECTION 2

Scalable Processing for Complex Data

- Distributed graph, hypergraph, and temporal analytics
- Hardware-aware vector and graph indexing
- Adaptive parallel runtimes for evolving workloads
- Emerging architectures for scalable execution

DIRECTION 3

Responsible Data Intelligence

- Industry collaboration in critical domains
- ESG, finance, law, mining, energy, and built environment
- Fairness, diversity, and accountable data use
- Climate, health, and scientific data intelligence

INDEXING

OPTIMIZATION

RUNTIME SYSTEMS

DISTRIBUTED
PROCESSING

DATA QUALITY

PROVENANCE

BENCHMARKS

Collaboration & Discussion

RESEARCH SUMMARY

Scalable Data Systems

Graph and hypergraph analytics, database runtime, streaming, and hardware-aware processing.

Responsible Data Intelligence

GraphRAG, LLM retrieval and reasoning, AI4DB, tabular learning, and data-centric AI.

Data for Real-World Applications

Industry collaboration, critical domains, fairness, climate, and health.

POSSIBLE COLLABORATIONS

RDBMS and HTAP systems

OLTP/HTAP engines, query execution, isolation and consistency, runtime optimization, and provenance.

Data engineering for AI and science

ML-in-DB, multicore/cloud execution, scientific data management, caching, and benchmarking.

Graph data management at scale

Higher-order and temporal graphs, subgraph matching, dynamic graph processing, and GraphRAG indexing.

Thank you!

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